Decimals Worksheet 1 – 1st ESO

Topic: Understanding Decimals and Place Value
Level: 1st Year Secondary Education (1st ESO)

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1. Theoretical Summary

Decimals are numbers that have a whole part and a fractional part separated by a decimal point. The place value of digits after the decimal point is tenths, hundredths, thousandths, and so on. For example, in 87.56, 5 is in the tenths place and 6 in the hundredths place.

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2. Exercises

1. Write these numbers in words:
   - 87.5
   - 90.25
   - 88.75
2. Identify the place value of the underlined digit:
   - 8**7**.56
   - 90.2**5**
   - 8**8**.75
3. Write these decimals in expanded form:
   - 87.5 = 80 + 7 + 0.5
   - 90.25 = 90 + 0 + 0.25
   - 88.75 = 80 + 8 + 0.75
4. Which decimal is bigger? 87.5 or 87.56? Explain why.
5. Convert the decimal numbers to fractions:
   - 0.5
   - 0.25
   - 0.75

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3. Competency Problems

1. Your radio is tuned to 87.5 MHz. If the frequency changes to 87.56 MHz, how much does the frequency increase by?
2. If you hear a station at 90.25 MHz, which has more whole units, 90 or 25 hundredths?

3. Explain why understanding place values is important for programming an FM radio.
4. If your radio only shows one decimal place, what would 88.75 MHz be rounded to?
5. How do fractions and decimals relate when tuning frequencies?

Solutions

Exercises:

1.
   - 87.5 → Eighty-seven point five
   - 90.25 → Ninety point two five
   - 88.75 → Eighty-eight point seven five
2.
   - In 87.56, 7 is in the units place
   - In 90.25, 5 is in the hundredths place
   - In 88.75, 8 is in the units place
3.
   - 87.5 = 80 + 7 + 0.5
   - 90.25 = 90 + 0 + 0.25
   - 88.75 = 80 + 8 + 0.75
4. 87.56 is bigger than 87.5 because it has more hundredths (0.06 more).
5.
   - 0.5 = 1/2
   - 0.25 = 1/4
   - 0.75 = 3/4

Competency Problems:

1. Increase = 87.56 - 87.5 = 0.06 MHz
2. Whole units: 90; fractional units: 25 hundredths
3. Place values help set the precise frequency in the radio's memory
4. 88.75 MHz rounded to one decimal place is 88.8 MHz
5. Fractions and decimals are two ways to represent parts of a whole, useful for tuning exact frequencies.
